Abstract:

As Large Language Models (LLMs) increasingly transition from isolated coding assistants to autonomous, interacting entities, software engineering faces a critical governance challenge. While purely neural multi-agent systems exhibit remarkable capabilities in natural language understanding, generative planning, and probabilistic code generation, they fundamentally lack the deterministic reliability, formal explainability, and rigorous safety guarantees required for mission-critical deployments. To address this inherent limitation, the paradigm of Neuro-Symbolic Artificial Intelligence (NSAI) offers a vital synthesis, combining the adaptable pattern recognition of neural networks with the rigorous, rule-based logic of symbolic reasoners. This paper explores the emerging domain of Architectural Patterns for Neuro-Symbolic Multi-Agent Systems, aiming to formalize how these highly complex hybrid systems should be systematically engineered and governed.

We conduct a comprehensive analysis of contemporary NSAI architectural models, focusing on integration frameworks specifically designed for multi-agent coordination . Drawing upon recent prototypical architectures, such as the Software Architecture for LLM-Based Multi-Agent Systems (SALLMA) , we identify and formalize compositional design patterns that safely dictate the interaction between probabilistic LLM agents and deterministic symbolic rule engines. Key architectural components examined include asynchronous communication protocols for robust agent coordination , hierarchical memory models that separate explicit symbolic states from ongoing agent discourse, and centralized orchestration mechanisms designed to manage computational workloads and strictly constrain agent autonomy.

Furthermore, this research evaluates specific configuration pipelines, such as the Neuro-Symbolic-Neuro pattern, which has demonstrated superior performance in aligning multi-agent systems with strict operational constraints and advanced technologies like retrieval-augmented generation. By formalizing these complex interactions into reusable software architecture patterns, we address fundamental engineering challenges regarding seamless system integration, computational efficiency, and the prevention of agentic privilege escalation . Ultimately, this work provides system architects with a standardized vocabulary and robust design frameworks, laying the necessary groundwork for deploying scalable, interpretable, and safe neuro-symbolic multi-agent systems in real-world environments